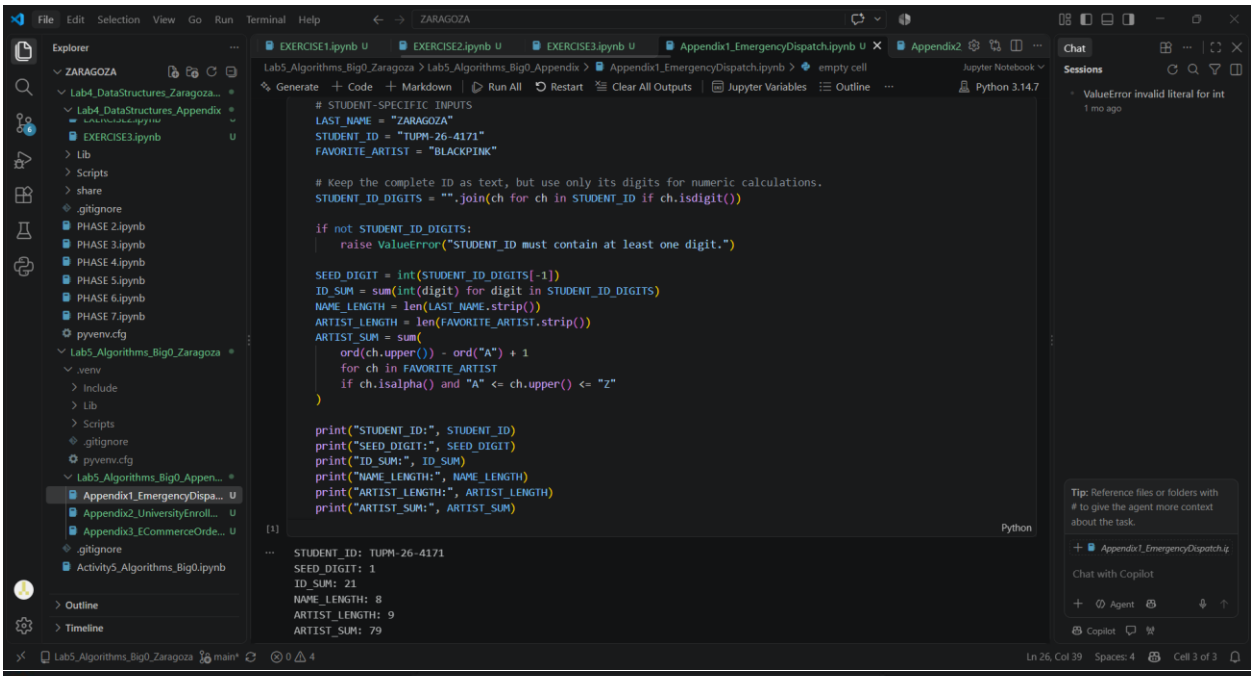
Zaragoza, Samson Jr. C. **Computer Programming**

BSECE – 1C **Activity 5: Appendix**

EXERCISE 1: Emergency Dispatch Record Search



```
# STUDENT-SPECIFIC INPUTS
LAST_NAME = "ZARAGOZA"
STUDENT_ID = "TUPM-26-4171"
FAVORITE_ARTIST = "BLACKPINK"

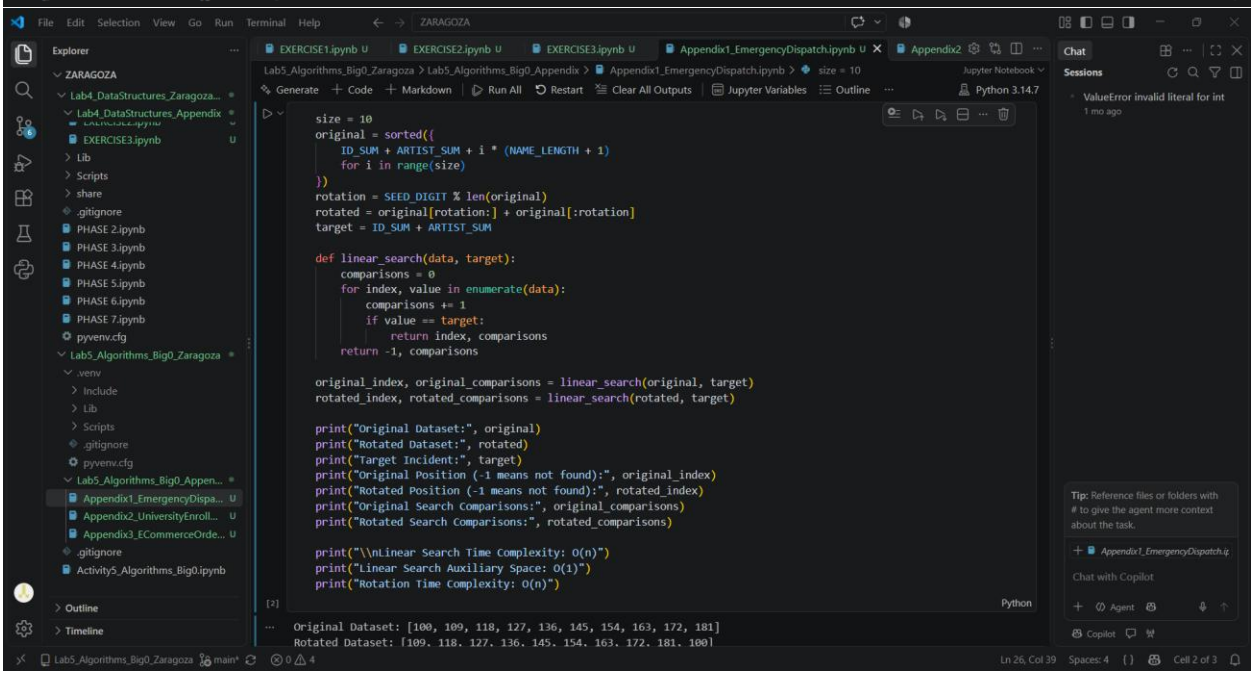
# Keep the complete ID as text, but use only its digits for numeric calculations.
STUDENT_ID_DIGITS = "".join(ch for ch in STUDENT_ID if ch.isdigit())

if not STUDENT_ID_DIGITS:
    raise ValueError("STUDENT_ID must contain at least one digit.")

SEED_DIGIT = int(STUDENT_ID_DIGITS[-1])
ID_SUM = sum(int(digit) for digit in STUDENT_ID_DIGITS)
NAME_LENGTH = len(LAST_NAME.strip())
ARTIST_LENGTH = len(FAVORITE_ARTIST.strip())
ARTIST_SUM = sum(
    ord(ch.upper()) - ord("A") + 1
    for ch in FAVORITE_ARTIST
    if ch.isalpha() and "A" <= ch.upper() <= "Z"
)

print("STUDENT_ID:", STUDENT_ID)
print("SEED_DIGIT:", SEED_DIGIT)
print("ID_SUM:", ID_SUM)
print("NAME_LENGTH:", NAME_LENGTH)
print("ARTIST_LENGTH:", ARTIST_LENGTH)
print("ARTIST_SUM:", ARTIST_SUM)
```

```
STUDENT_ID: TUPM-26-4171
SEED_DIGIT: 1
ID_SUM: 21
NAME_LENGTH: 8
ARTIST_LENGTH: 9
ARTIST_SUM: 79
```



```
size = 10
original = sorted({
    ID_SUM + ARTIST_SUM + i * (NAME_LENGTH + 1)
    for i in range(size)
})
rotation = SEED_DIGIT % len(original)
rotated = original[rotation:] + original[:rotation]
target = ID_SUM + ARTIST_SUM

def linear_search(data, target):
    comparisons = 0
    for index, value in enumerate(data):
        comparisons += 1
        if value == target:
            return index, comparisons
    return -1, comparisons

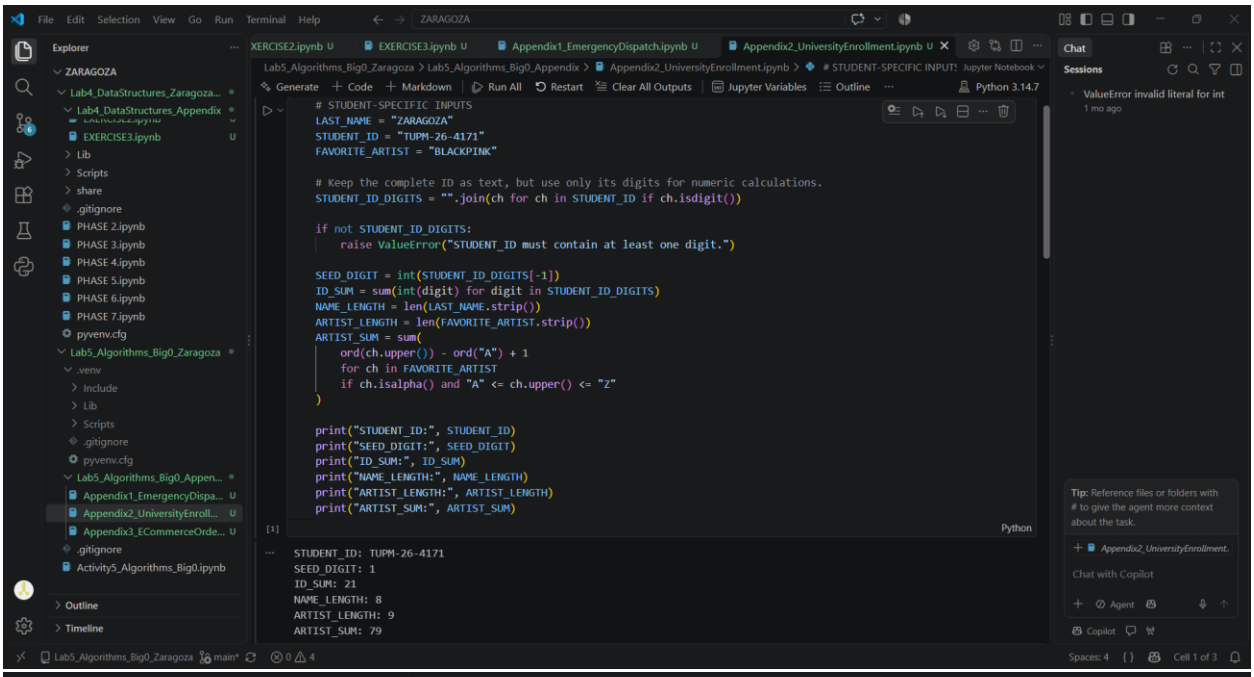
original_index, original_comparisons = linear_search(original, target)
rotated_index, rotated_comparisons = linear_search(rotated, target)

print("Original Dataset:", original)
print("Rotated Dataset:", rotated)
print("Target Incident:", target)
print("Original Position (-1 means not found):", original_index)
print("Rotated Position (-1 means not found):", rotated_index)
print("Original Search Comparisons:", original_comparisons)
print("Rotated Search Comparisons:", rotated_comparisons)

print("\\nLinear Search Time Complexity: O(n)")
print("Linear Search Auxiliary Space: O(1)")
print("Rotation Time Complexity: O(n)")
```

```
Original Dataset: [100, 109, 118, 127, 136, 145, 154, 163, 172, 181]
Rotated Dataset: [109, 118, 127, 136, 145, 154, 163, 172, 181, 100]
```

EXERCISE 2: University Enrollment Records



```
# STUDENT-SPECIFIC INPUTS
LAST_NAME = "ZARAGOZA"
STUDENT_ID = "TUPM-26-4171"
FAVORITE_ARTIST = "BLACKPINK"

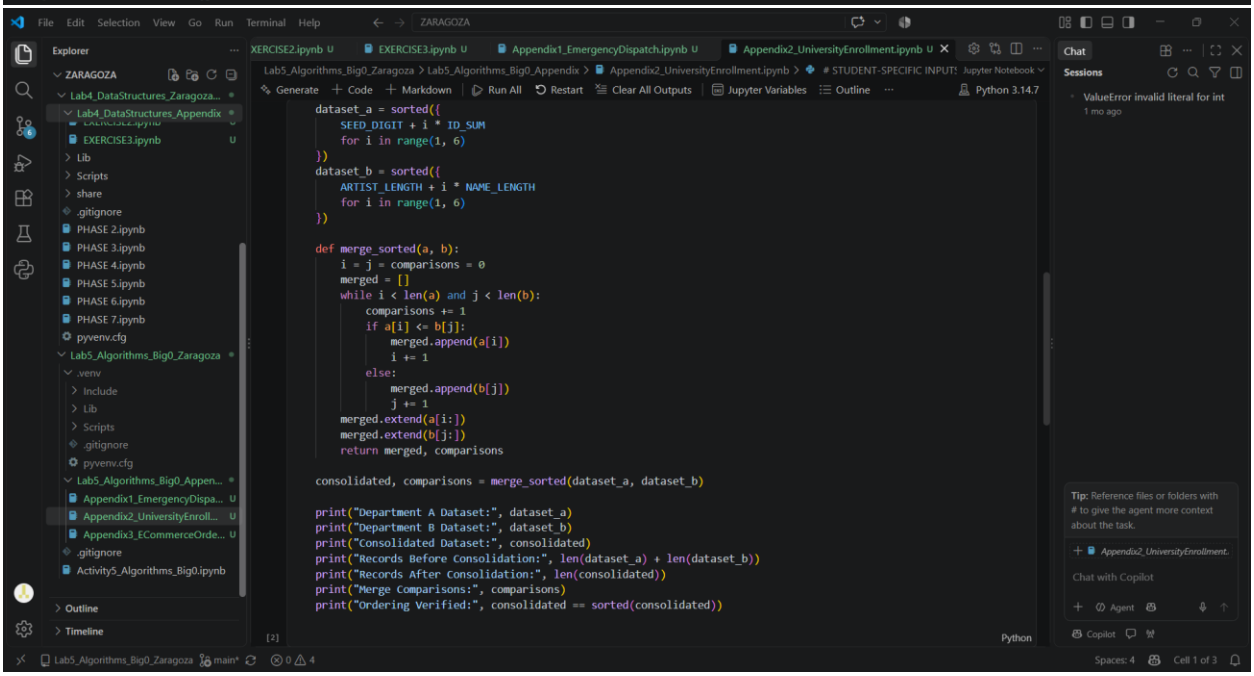
# Keep the complete ID as text, but use only its digits for numeric calculations.
STUDENT_ID_DIGITS = "".join(ch for ch in STUDENT_ID if ch.isdigit())

if not STUDENT_ID_DIGITS:
    raise ValueError("STUDENT_ID must contain at least one digit.")

SEED_DIGIT = int(STUDENT_ID_DIGITS[-1])
ID_SUM = sum(int(digit) for digit in STUDENT_ID_DIGITS)
NAME_LENGTH = len(LAST_NAME.strip())
ARTIST_LENGTH = len(FAVORITE_ARTIST.strip())
ARTIST_SUM = sum(
    ord(ch.upper()) - ord("A") + 1
    for ch in FAVORITE_ARTIST
    if ch.isalpha() and "A" <= ch.upper() <= "Z"
)

print("STUDENT_ID:", STUDENT_ID)
print("SEED_DIGIT:", SEED_DIGIT)
print("ID_SUM:", ID_SUM)
print("NAME_LENGTH:", NAME_LENGTH)
print("ARTIST_LENGTH:", ARTIST_LENGTH)
print("ARTIST_SUM:", ARTIST_SUM)
```

```
STUDENT_ID: TUPM-26-4171
SEED_DIGIT: 1
ID_SUM: 21
NAME_LENGTH: 8
ARTIST_LENGTH: 9
ARTIST_SUM: 79
```

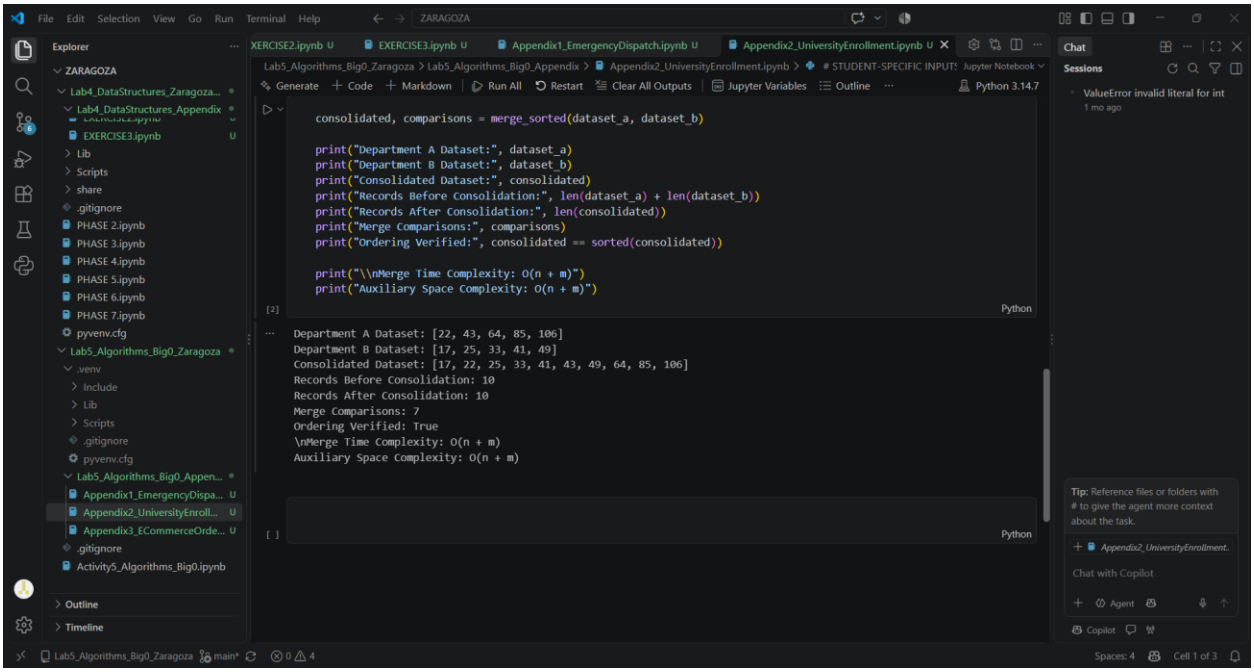


```
dataset_a = sorted({
    SEED_DIGIT + i * ID_SUM
    for i in range(1, 6)
})
dataset_b = sorted({
    ARTIST_LENGTH + i * NAME_LENGTH
    for i in range(1, 6)
})

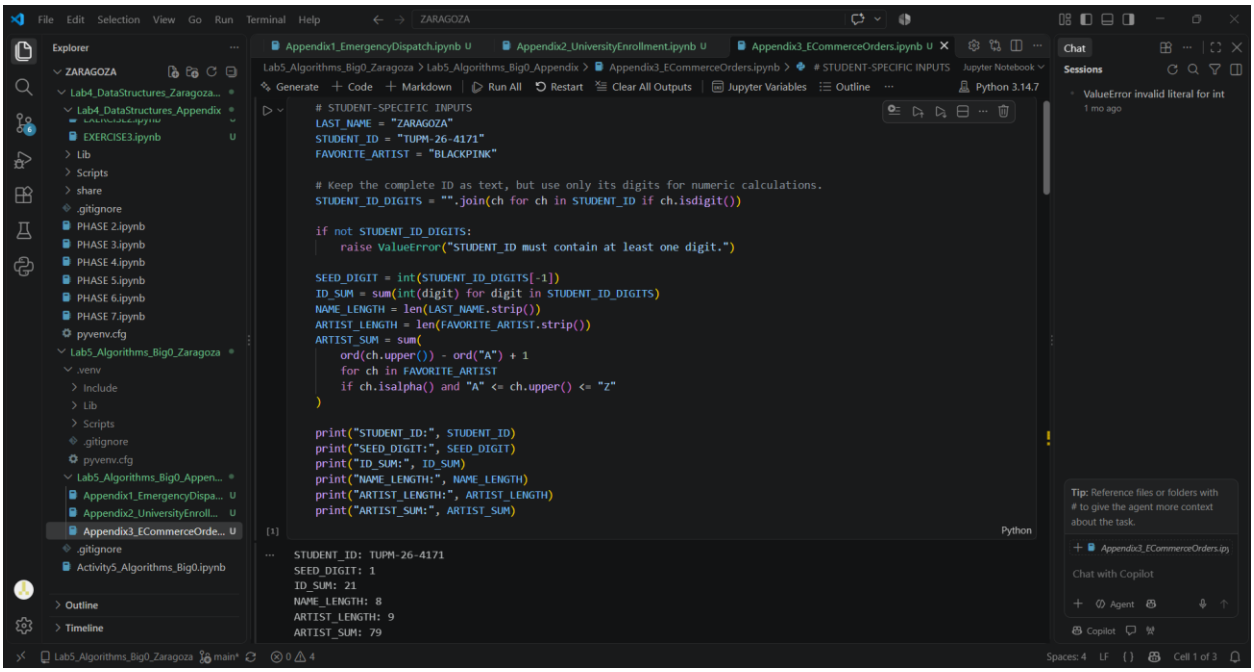
def merge_sorted(a, b):
    i = j = comparisons = 0
    merged = []
    while i < len(a) and j < len(b):
        comparisons += 1
        if a[i] <= b[j]:
            merged.append(a[i])
            i += 1
        else:
            merged.append(b[j])
            j += 1
    merged.extend(a[i:])
    merged.extend(b[j:])
    return merged, comparisons

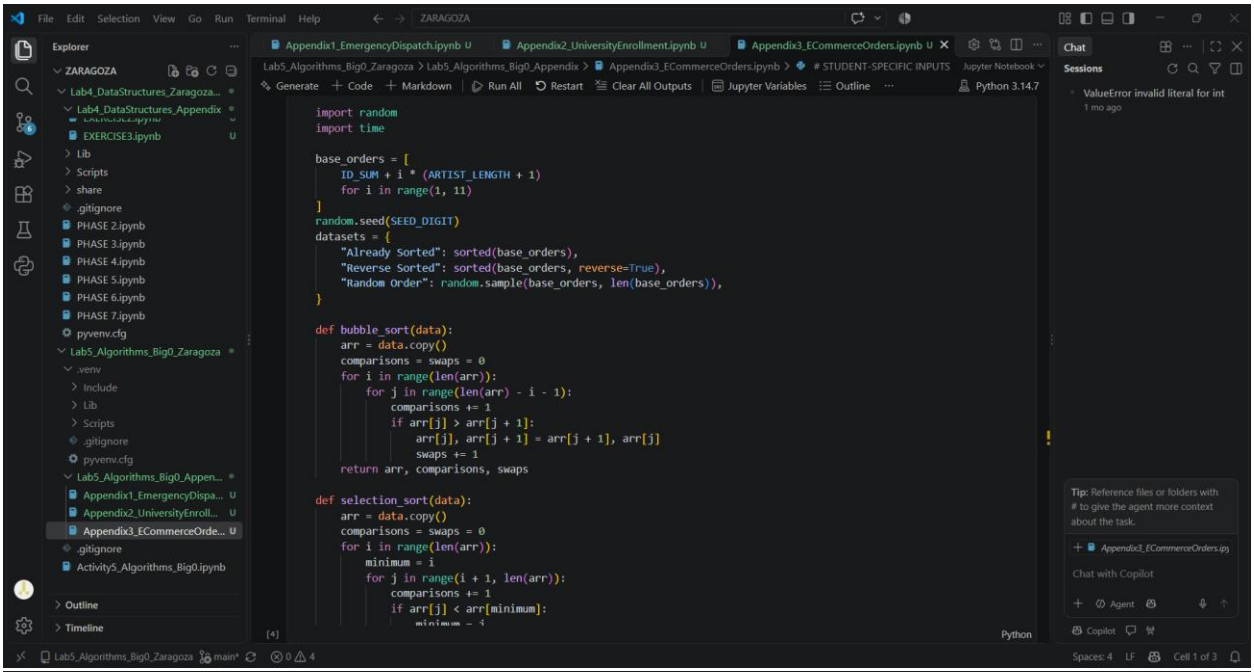
consolidated, comparisons = merge_sorted(dataset_a, dataset_b)

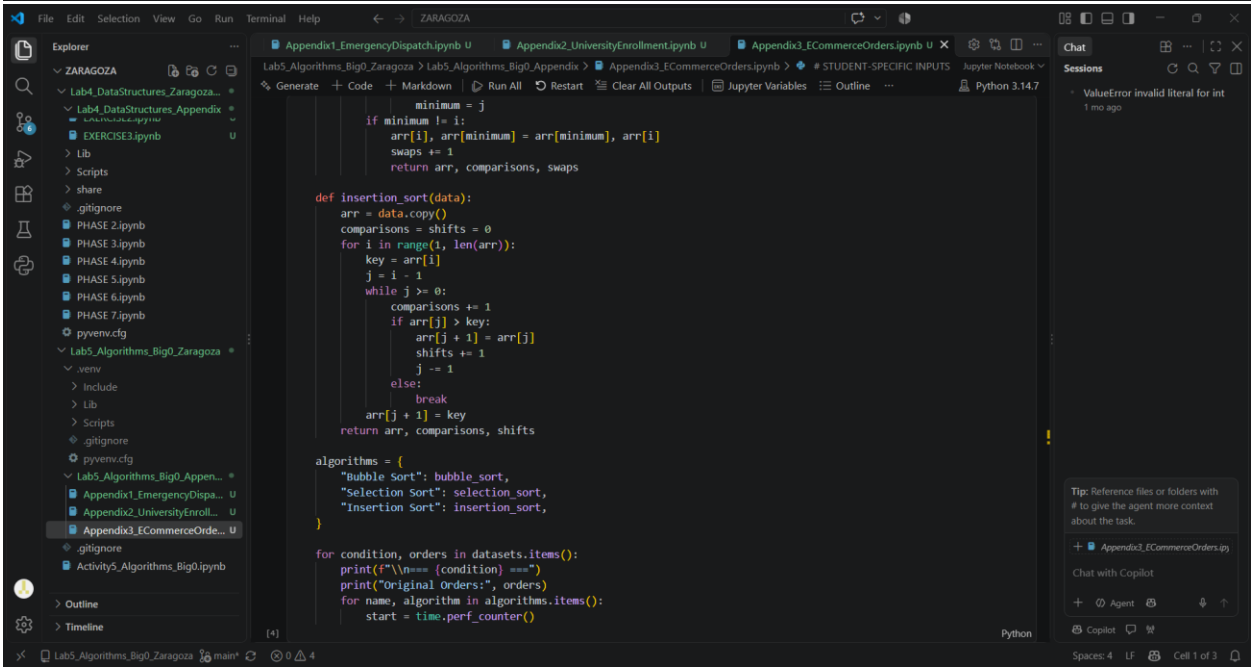
print("Department A Dataset:", dataset_a)
print("Department B Dataset:", dataset_b)
print("Consolidated Dataset:", consolidated)
print("Records Before Consolidation:", len(dataset_a) + len(dataset_b))
print("Records After Consolidation:", len(consolidated))
print("Merge Comparisons:", comparisons)
print("Ordering Verified:", consolidated == sorted(consolidated))
```

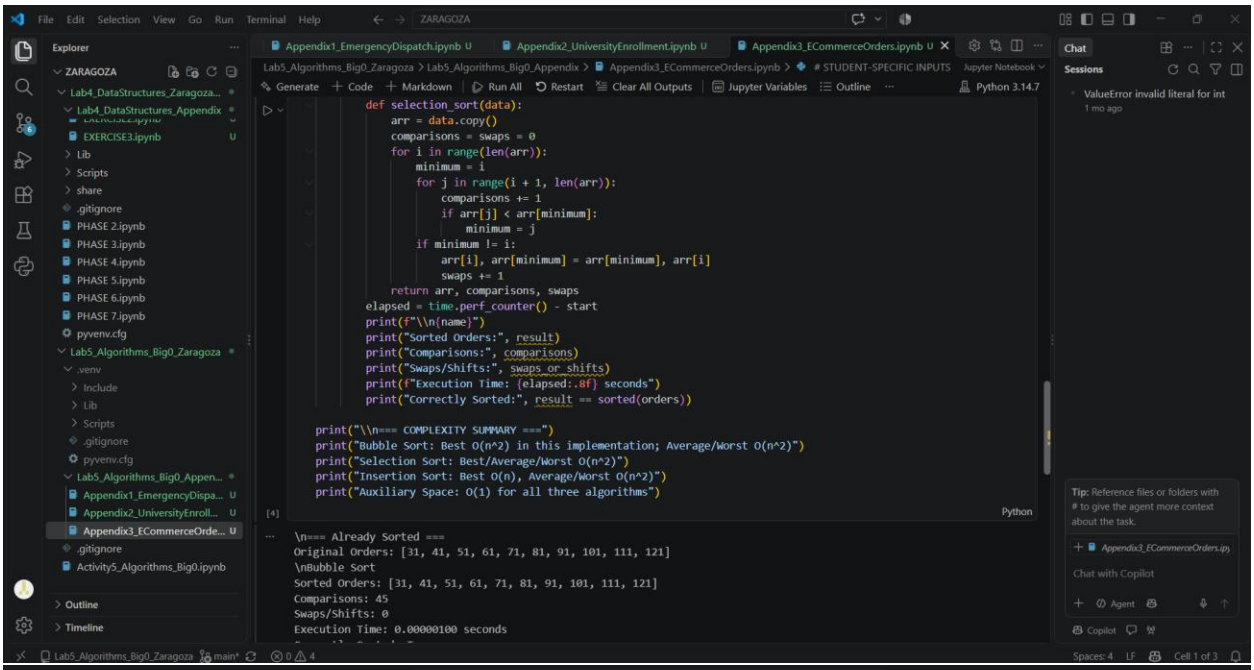


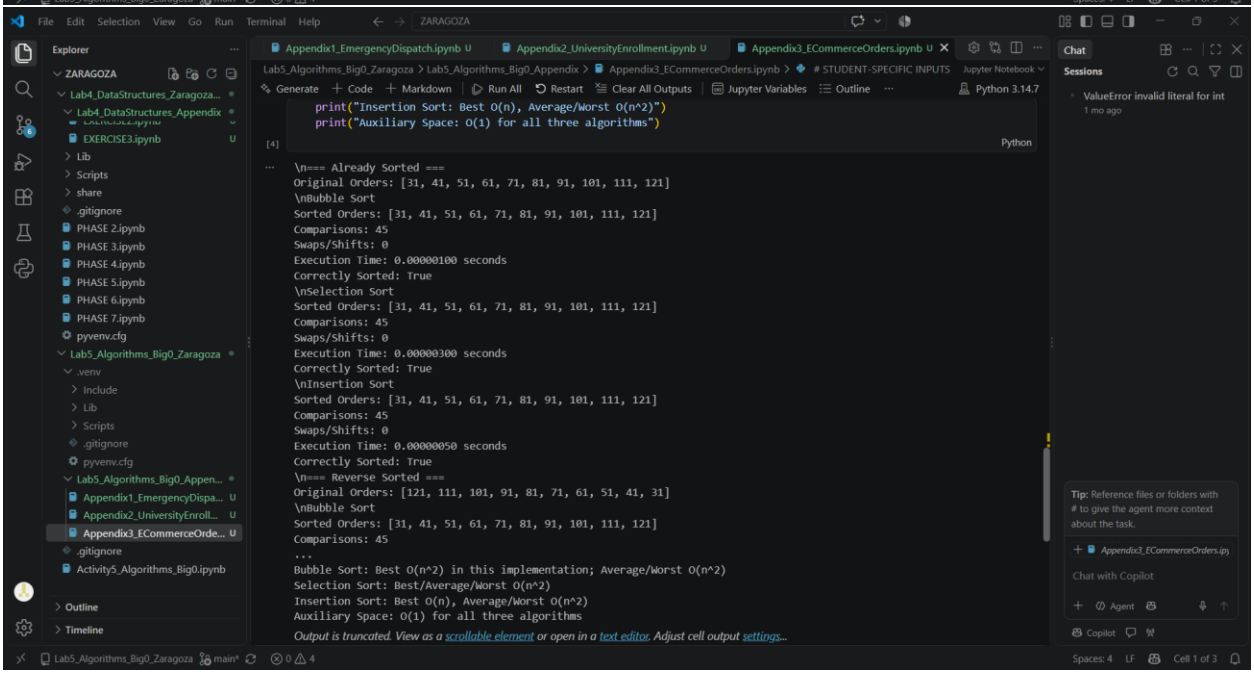
EXERCISE 3: E-Commerce Order Processing




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ZARAGOZA
Explorer
ZARAGOZA
Lab4_DataStructures_Zaragoza...
Lab4_DataStructures_Appendix
EXERCISE3.ipynb
Lib
Scripts
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.gitignore
PHASE 2.ipynb
PHASE 3.ipynb
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PHASE 5.ipynb
PHASE 6.ipynb
PHASE 7.ipynb
pyvenv.cfg
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Appendix2_UniversityEnroll... U
Appendix3_ECommerceOrde... U
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Appendix2_UniversityEnrollment.ipynb U
Appendix3_ECommerceOrders.ipynb U
Lab5_Algorithms_Big0_Zaragoza > Lab5_Algorithms_Big0_Appendix > Appendix3_ECommerceOrders.ipynb > # STUDENT-SPECIFIC INPUTS
Jupyter Notebook
Generate Code Markdown Run All Restart Clear All Outputs Jupyter Variables Outline
Python 3.14.7
import random
import time
base_orders = [
ID_SUM + i * (ARTIST_LENGTH + 1)
for i in range(1, 11)
]
random.seed(SEED_DIGIT)
datasets = {
"Already Sorted": sorted(base_orders),
"Reverse Sorted": sorted(base_orders, reverse=True),
"Random Order": random.sample(base_orders, len(base_orders)),
}
def bubble_sort(data):
arr = data.copy()
comparisons = swaps = 0
for i in range(len(arr)):
for j in range(len(arr) - i - 1):
comparisons += 1
if arr[j] > arr[j + 1]:
arr[j], arr[j + 1] = arr[j + 1], arr[j]
swaps += 1
return arr, comparisons, swaps
def selection_sort(data):
arr = data.copy()
comparisons = swaps = 0
for i in range(len(arr)):
minimum = i
for j in range(i + 1, len(arr)):
comparisons += 1
if arr[j] < arr[minimum]:
[4]
Python
Chat
Sessions
ValueError invalid literal for int
1 mo ago
Tip: Reference files or folders with # to give the agent more context about the task.
Appendix3_ECommerceOrders.ipy
Chat with Copilot
Agent
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PHASE 4.ipynb
PHASE 5.ipynb
PHASE 6.ipynb
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Appendix2_UniversityEnroll... U
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Appendix3_ECommerceOrders.ipynb U
Lab5_Algorithms_Big0_Zaragoza > Lab5_Algorithms_Big0_Appendix > Appendix3_ECommerceOrders.ipynb > # STUDENT-SPECIFIC INPUTS
Jupyter Notebook
Generate Code Markdown Run All Restart Clear All Outputs Jupyter Variables Outline
Python 3.14.7
minimum = j
if minimum != i:
arr[i], arr[minimum] = arr[minimum], arr[i]
swaps += 1
return arr, comparisons, swaps
def insertion_sort(data):
arr = data.copy()
comparisons = shifts = 0
for i in range(1, len(arr)):
key = arr[i]
j = i - 1
while j >= 0:
comparisons += 1
if arr[j] > key:
arr[j + 1] = arr[j]
shifts += 1
j -= 1
else:
break
arr[j + 1] = key
return arr, comparisons, shifts
algorithms = {
"Bubble Sort": bubble_sort,
"Selection Sort": selection_sort,
"Insertion Sort": insertion_sort,
}
for condition, orders in datasets.items():
print(f"\\n=== {condition} ===")
print("Original Orders:", orders)
for name, algorithm in algorithms.items():
start = time.perf_counter()
[4]
Python
Chat
Sessions
ValueError invalid literal for int
1 mo ago
Tip: Reference files or folders with # to give the agent more context about the task.
Appendix3_ECommerceOrders.ipy
Chat with Copilot
Agent
Copilot
Lab5_Algorithms_Big0_Zaragoza main*
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Spaces: 4 LF Cell 1 of 3


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Appendix2_UniversityEnrollment.ipynb U
Appendix3_ECommerceOrders.ipynb U
Lab5_Algorithms_Big0_Zaragoza > Lab5_Algorithms_Big0_Appendix > Appendix3_ECommerceOrders.ipynb > # STUDENT-SPECIFIC INPUTS
Jupyter Notebook
Generate Code Markdown Run All Restart Clear All Outputs Jupyter Variables Outline
Python 3.14.7
def selection_sort(data):
arr = data.copy()
comparisons = swaps = 0
for i in range(len(arr)):
minimum = i
for j in range(i + 1, len(arr)):
comparisons += 1
if arr[j] < arr[minimum]:
minimum = j
if minimum != i:
arr[i], arr[minimum] = arr[minimum], arr[i]
swaps += 1
return arr, comparisons, swaps
elapsed = time.perf_counter() - start
print(f"\\n{name}")
print("Sorted Orders:", result)
print("Comparisons:", comparisons)
print("Swaps/Shifts:", swaps_or_shifts)
print(f"Execution Time: {elapsed:.8f} seconds")
print("Correctly Sorted:", result == sorted(orders))
print("\\n=== COMPLEXITY SUMMARY ===")
print("Bubble Sort: Best O(n^2) in this implementation; Average/Worst O(n^2)")
print("Selection Sort: Best/Average/Worst O(n^2)")
print("Insertion Sort: Best O(n), Average/Worst O(n^2)")
print("Auxiliary Space: O(1) for all three algorithms")
[4]
Python
\n=== Already Sorted ===
Original Orders: [31, 41, 51, 61, 71, 81, 91, 101, 111, 121]
\nBubble Sort
Sorted Orders: [31, 41, 51, 61, 71, 81, 91, 101, 111, 121]
Comparisons: 45
Swaps/Shifts: 0
Execution Time: 0.00000100 seconds
Chat
Sessions
ValueError invalid literal for int
1 mo ago
Tip: Reference files or folders with # to give the agent more context about the task.
Chat with Copilot
Agent
Copilot
Lab5_Algorithms_Big0_Zaragoza main*
Spaces: 4 LF Cell 1 of 3


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Appendix1_EmergencyDispatch.ipynb U
Appendix2_UniversityEnrollment.ipynb U
Appendix3_ECommerceOrders.ipynb U
Lab5_Algorithms_Big0_Zaragoza > Lab5_Algorithms_Big0_Appendix > Appendix3_ECommerceOrders.ipynb > # STUDENT-SPECIFIC INPUTS
Jupyter Notebook
Generate Code Markdown Run All Restart Clear All Outputs Jupyter Variables Outline
Python 3.14.7
print("Insertion Sort: Best O(n), Average/Worst O(n^2)")
print("Auxiliary Space: O(1) for all three algorithms")
[4]
Python
\n=== Already Sorted ===
Original Orders: [31, 41, 51, 61, 71, 81, 91, 101, 111, 121]
\nBubble Sort
Sorted Orders: [31, 41, 51, 61, 71, 81, 91, 101, 111, 121]
Comparisons: 45
Swaps/Shifts: 0
Execution Time: 0.00000100 seconds
Correctly Sorted: True
\nSelection Sort
Sorted Orders: [31, 41, 51, 61, 71, 81, 91, 101, 111, 121]
Comparisons: 45
Swaps/Shifts: 0
Execution Time: 0.00000300 seconds
Correctly Sorted: True
\nInsertion Sort
Sorted Orders: [31, 41, 51, 61, 71, 81, 91, 101, 111, 121]
Comparisons: 45
Swaps/Shifts: 0
Execution Time: 0.00000050 seconds
Correctly Sorted: True
\n=== Reverse Sorted ===
Original Orders: [121, 111, 101, 91, 81, 71, 61, 51, 41, 31]
\nBubble Sort
Sorted Orders: [31, 41, 51, 61, 71, 81, 91, 101, 111, 121]
Comparisons: 45
...
Bubble Sort: Best O(n^2) in this implementation; Average/Worst O(n^2)
Selection Sort: Best/Average/Worst O(n^2)
Insertion Sort: Best O(n), Average/Worst O(n^2)
Auxiliary Space: O(1) for all three algorithms
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
Chat
Sessions
ValueError invalid literal for int
1 mo ago
Tip: Reference files or folders with # to give the agent more context about the task.
Chat with Copilot
Agent
Copilot
Lab5_Algorithms_Big0_Zaragoza main*
Spaces: 4 LF Cell 1 of 3